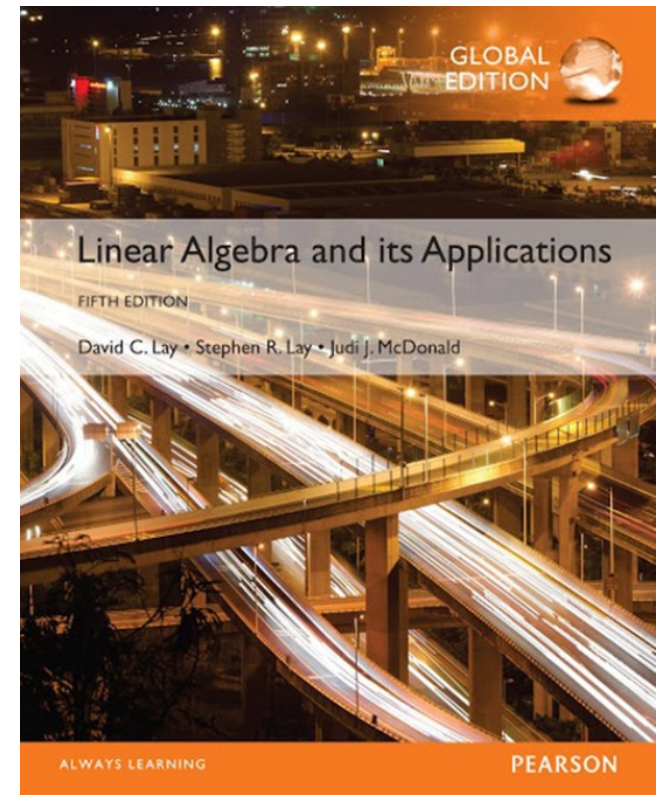


2

Matrix Algebra

2.1

MATRIX OPERATIONS



MATRIX OPERATIONS

- If A is an $m \times n$ matrix—that is, a matrix with m rows and n columns—then the scalar entry in the i th row and j th column of A is denoted by a_{ij} and is called the (i, j) -entry of A . See the Fig. 1 below.
- Each column of A is a list of m real numbers, which identifies a vector in $\mathbb{R}^m$.

The diagram shows a matrix A with m rows and n columns. The entry a_{ij} is highlighted in blue. The row i is labeled "Row i " and the column j is labeled "Column j ". Below the matrix, the column vectors $\mathbf{a}_1$, $\mathbf{a}_j$, and $\mathbf{a}_n$ are indicated by blue arrows pointing to the first, j th, and n th columns respectively.

$$\begin{array}{c}
 \text{Column } j \\
 \begin{bmatrix}
 a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\
 \vdots & & \vdots & & \vdots \\
 \text{Row } i & a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\
 \vdots & & \vdots & & \vdots \\
 a_{m1} & \cdots & a_{mj} & \cdots & a_{mn}
 \end{bmatrix} = A
 \end{array}$$

$\uparrow$ $\uparrow$ $\uparrow$
 $\mathbf{a}_1$ $\mathbf{a}_j$ $\mathbf{a}_n$

Matrix notation.

MATRIX OPERATIONS

- The columns are denoted by $\mathbf{a}_1, \dots, \mathbf{a}_n$, and the matrix A is written as

$$A = [a_1 \ a_2 \ \dots \ a_n]$$

- The number a_{ij} is the i th entry (from the top) of the j th column vector $\mathbf{a}_j$.
- The **diagonal entries** in an $m \times n$ matrix $A = [a_{ij}]$ are $a_{11}, a_{22}, a_{33}, \dots$, and they form the **main diagonal** of A .
- A **diagonal matrix** is a square $n \times n$ matrix whose nondiagonal entries are zero.
- An example is the $n \times n$ identity matrix, I_n .

SUMS AND SCALAR MULTIPLES

- An $m \times n$ matrix whose entries are all zero is a **zero matrix** and is written as 0 .
- The two matrices are **equal** if they have the same size (*i.e.*, the same number of rows and the same number of columns) and if their corresponding columns are equal, which amounts to saying that their corresponding entries are equal.
- If A and B are $m \times n$ matrices, then the **sum** $A + B$ is the $m \times n$ matrix whose columns are the sums of the corresponding columns in A and B .

SUMS AND SCALAR MULTIPLES

- Since vector addition of the columns is done entrywise, each entry in $A + B$ is the sum of the corresponding entries in A and B .
- The sum $A + B$ is defined only when A and B are the same size.
- **Example 1:** Let $A = \begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix}$,

and $C = \begin{bmatrix} 2 & -3 \\ 0 & 1 \end{bmatrix}$. Find $A + B$ and $A + C$.

SUMS AND SCALAR MULTIPLES

- **Solution:** $A + B = \begin{bmatrix} 5 & 1 & 6 \\ 2 & 8 & 9 \end{bmatrix}$ but $A + C$ is not defined because A and C have different sizes.
- If r is a scalar and A is a matrix, then the **scalar multiple** rA is the matrix whose columns are r times the corresponding columns in A .
- **Theorem 1:** Let A , B , and C be matrices of the same size, and let r and s be scalars.
 - a. $A + B = B + A$

SUMS AND SCALAR MULTIPLES

b. $(A + B) + C = A + (B + C)$

c. $A + 0 = A$

d. $r(A + B) = rA + rB$

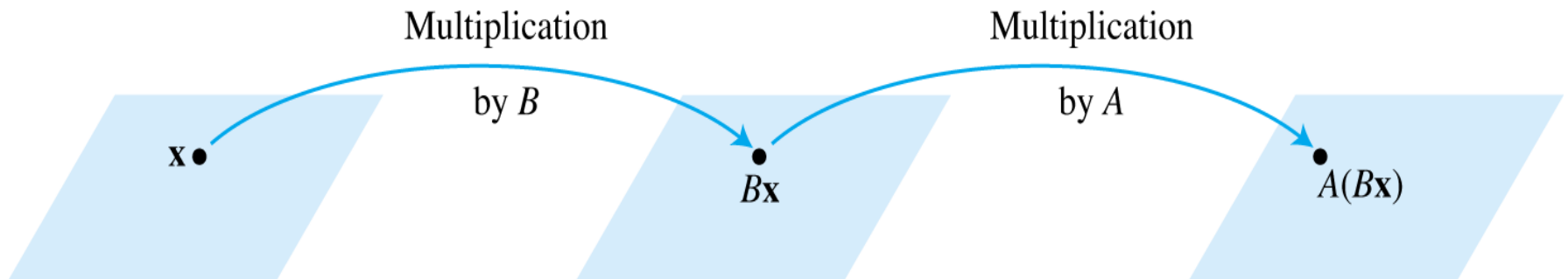
e. $(r + s)A = rA + sA$

f. $r(sA) = (rs)A$

- Each quantity in Theorem 1 is verified by showing that the matrix on the left side has the same size as the matrix on the right and that corresponding columns are equal.

MATRIX MULTIPLICATION

- When a matrix B multiplies a vector $\mathbf{x}$, it transforms $\mathbf{x}$ into the vector $B\mathbf{x}$.
- If this vector is then multiplied in turn by a matrix A , the resulting vector is $A(B\mathbf{x})$. See the Fig. 2 below.

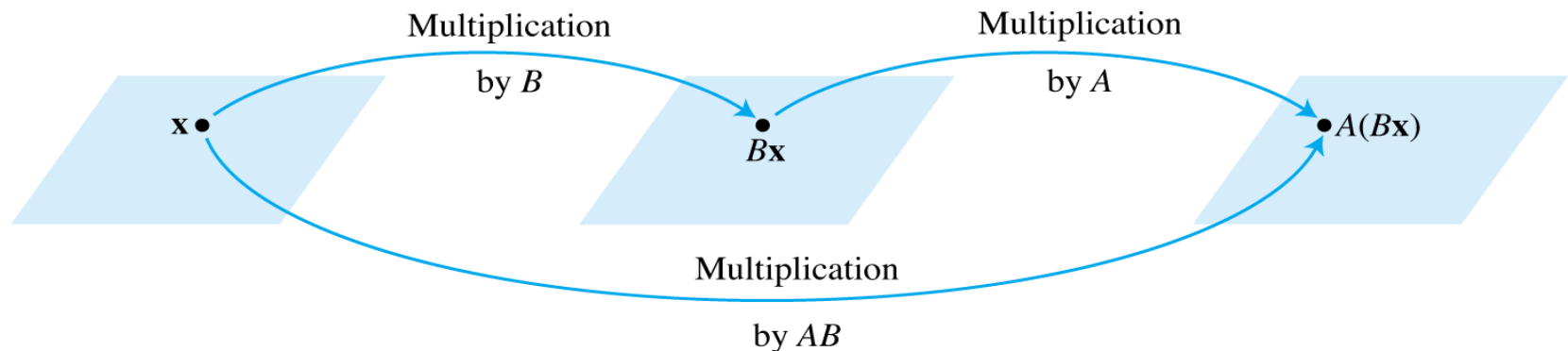


Multiplication by B and then A .

- Thus $A(B\mathbf{x})$ is produced from $\mathbf{x}$ by a *composition of mappings*—the linear transformations.

MATRIX MULTIPLICATION

- Our goal is to represent this composite mapping as multiplication by a single matrix, denoted by AB , so that $A(B\mathbf{x}) = (AB)\mathbf{x}$. See Fig. 3 below



Multiplication by AB .

- If A is $m \times n$, B is $n \times p$, and $\mathbf{x}$ is in $\mathbb{R}^p$, denote the columns of B by $\mathbf{b}_1, \dots, \mathbf{b}_p$ and the entries in $\mathbf{x}$ by $x_1, \dots, x_p$.

MATRIX MULTIPLICATION

- Then

$$B\mathbf{x} = x_1\mathbf{b}_1 + \dots + x_p\mathbf{b}_p$$

- By the linearity of multiplication by A ,

$$\begin{aligned} A(B\mathbf{x}) &= A(x_1\mathbf{b}_1) + \dots + A(x_p\mathbf{b}_p) \\ &= x_1A\mathbf{b}_1 + \dots + x_pA\mathbf{b}_p \end{aligned}$$

- The vector $A(B\mathbf{x})$ is a linear combination of the vectors $A\mathbf{b}_1, \dots, A\mathbf{b}_p$, using the entries in $\mathbf{x}$ as weights.
- In matrix notation, this linear combination is written as

$$A(B\mathbf{x}) = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \dots & A\mathbf{b}_p \end{bmatrix} \mathbf{x}$$

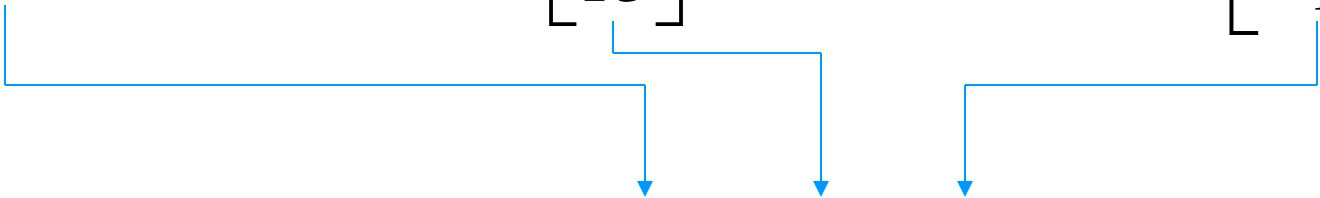
MATRIX MULTIPLICATION

- Thus multiplication by $\begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \end{bmatrix}$ transforms $\mathbf{x}$ into $A(B\mathbf{x})$.
- **Definition:** If A is an $m \times n$ matrix, and if B is an $n \times p$ matrix with columns $\mathbf{b}_1, \dots, \mathbf{b}_p$, then the product AB is the $m \times p$ matrix whose columns are $A\mathbf{b}_1, \dots, A\mathbf{b}_p$.
- That is,
$$AB = A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_p \end{bmatrix} = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \end{bmatrix}$$
- *Multiplication of matrices corresponds to composition of linear transformations.*

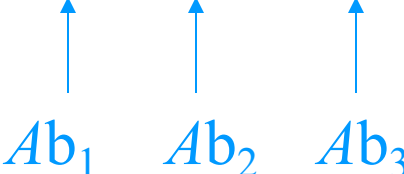
MATRIX MULTIPLICATION

- **Example 3:** Compute AB , where $A = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}$.
- **Solution:** Write $B = [b_1 \quad b_2 \quad b_3]$, and compute:

MATRIX MULTIPLICATION

$$Ab_1 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix}, Ab_2 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix}, Ab_3 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$
$$= \begin{bmatrix} 11 \\ -1 \end{bmatrix} \quad = \begin{bmatrix} 0 \\ 13 \end{bmatrix} \quad = \begin{bmatrix} 21 \\ -9 \end{bmatrix}$$


■ Then

$$AB = A[b_1 \quad b_2 \quad b_3] = \begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \end{bmatrix}$$


Ab_1 Ab_2 Ab_3

MATRIX MULTIPLICATION

- Each column of AB is a linear combination of the columns of A using weights from the corresponding column of B .

Row—column rule for computing AB

- If a product AB is defined, then the entry in row i and column j of AB is the sum of the products of corresponding entries from row i of A and column j of B . If $(AB)_{ij}$ denotes the (i, j) -entry in AB , and if A is an $m \times n$ matrix, then

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$$

PROPERTIES OF MATRIX MULTIPLICATION

- **Theorem 2:** Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined.
 - a. $A(BC) = (AB)C$ (associative law of multiplication)
 - b. $A(B + C) = AB + AC$ (left distributive law)
 - c. $(B + C)A = BA + CA$ (right distributive law)
 - d. $r(AB) = (rA)B = A(rB)$ for any scalar r
 - e. $I_m A = A = A I_n$ (identity for matrix multiplication)

PROPERTIES OF MATRIX MULTIPLICATION

- **Proof:** Property (a) follows from the fact that matrix multiplication corresponds to composition of linear transformations (which are functions), and it is known that the composition of functions is associative. Let

$$C = \begin{bmatrix} \mathbf{c}_1 & \cdots & \mathbf{c}_p \end{bmatrix}$$

- By the definition of matrix multiplication,

$$BC = \begin{bmatrix} B\mathbf{c}_1 & \cdots & B\mathbf{c}_p \end{bmatrix}$$

$$A(BC) = \begin{bmatrix} A(B\mathbf{c}_1) & \cdots & A(B\mathbf{c}_p) \end{bmatrix}$$

PROPERTIES OF MATRIX MULTIPLICATION

- The definition of AB makes $A(B\mathbf{x}) = (AB)\mathbf{x}$ for all $\mathbf{x}$, so

$$A(BC) = \begin{bmatrix} (AB)\mathbf{c}_1 & \cdots & (AB)\mathbf{c}_p \end{bmatrix} = (AB)C$$

- The left-to-right order in products is critical because AB and BA are usually not the same.
- Because the columns of AB are linear combinations of the columns of A , whereas the columns of BA are constructed from the columns of B .
- The position of the factors in the product AB is emphasized by saying that A is *right-multiplied* by B or that B is *left-multiplied* by A .

PROPERTIES OF MATRIX MULTIPLICATION

- If $AB = BA$, we say that A and B **commute** with one another.

- **Warnings:**
 1. In general, $AB \neq BA$.
 2. The cancellation laws do *not* hold for matrix multiplication. That is, if $AB = AC$, then it is *not* true in general that $B = C$.
 3. If a product AB is the zero matrix, you *cannot* conclude in general that either $A = 0$ or $B = 0$.

POWERS OF A MATRIX

- If A is an $n \times n$ matrix and if k is a positive integer, then A^k denotes the product of k copies of A :

$$A^k = \underbrace{A \cdots A}_k$$

- If A is nonzero and if $\mathbf{x}$ is in $\mathbb{R}^n$, then $A^k \mathbf{x}$ is the result of left-multiplying $\mathbf{x}$ by A repeatedly k times.
- If $k = 0$, then $A^0 \mathbf{x}$ should be $\mathbf{x}$ itself.
- Thus A^0 is interpreted as the identity matrix.

THE TRANSPOSE OF A MATRIX

- Given an $m \times n$ matrix A , the **transpose** of A is the $n \times m$ matrix, denoted by A^T , whose columns are formed from the corresponding rows of A .

Theorem 3: Let A and B denote matrices whose sizes are appropriate for the following sums and products.

a. $(A^T)^T = A$

b. $(A + B)^T = A^T + B^T$

c. For any scalar r , $(rA)^T = rA^T$

d. $(AB)^T = B^T A^T$

THE TRANSPOSE OF A MATRIX

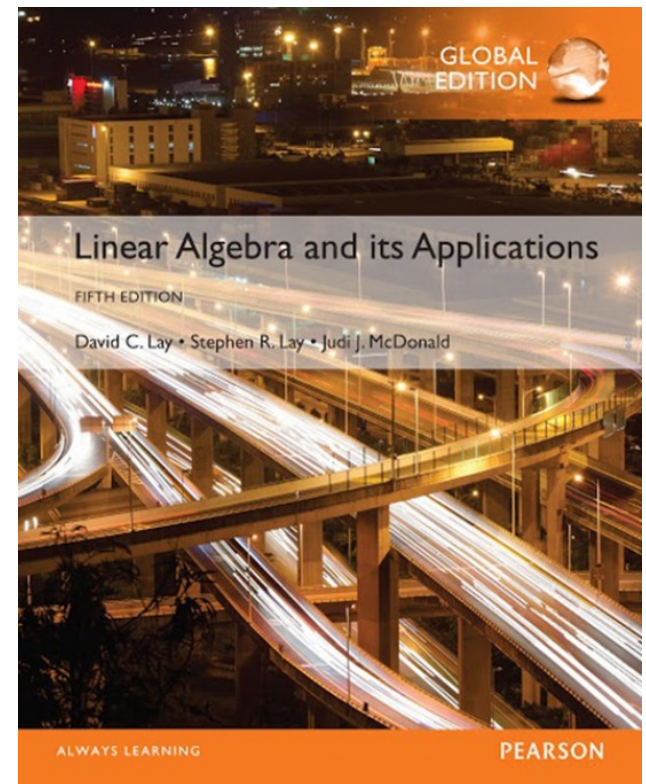
- The transpose of a product of matrices equals the product of their transposes in the *reverse* order.

2

Matrix Algebra

2.2

THE INVERSE OF A MATRIX



MATRIX OPERATIONS

- An $n \times n$ matrix A is said to be **invertible** if there is an $n \times n$ matrix C such that

$$CA = I \quad \text{and} \quad AC = I$$

where $I = I_n$, the $n \times n$ identity matrix.

- In this case, C is an **inverse** of A .
- In fact, C is uniquely determined by A , because if B were another inverse of A , then

$$B = BI = B(AC) = (BA)C = IC = C.$$

- This unique inverse is denoted by A^{-1} , so that

$$A^{-1}A = I \quad \text{and} \quad AA^{-1} = I.$$

MATRIX OPERATIONS

- **Theorem 4:** Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then

A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - bc = 0$, then A is not invertible.

- The quantity $ad - bc$ is called the **determinant** of A , and we write $\det A = ad - bc$
- This theorem says that a 2×2 matrix A is invertible if and only if $\det A \neq 0$

MATRIX OPERATIONS

- **Theorem 5:** If A is an invertible $n \times n$ matrix, then for each $\mathbf{b}$ in $\mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.
- **Proof:** Take any $\mathbf{b}$ in $\mathbb{R}^n$.
- A solution exists because if $A^{-1}\mathbf{b}$ is substituted for $\mathbf{x}$, then $A\mathbf{x} = A(A^{-1}\mathbf{b}) = (AA^{-1})\mathbf{b} = I\mathbf{b} = \mathbf{b}$.
- So $A^{-1}\mathbf{b}$ is a solution.
- To prove that the solution is unique, show that if $\mathbf{u}$ is any solution, then $\mathbf{u}$ must be $A^{-1}\mathbf{b}$.
- If $A\mathbf{u} = \mathbf{b}$, we can multiply both sides by A^{-1} and obtain $A^{-1}A\mathbf{u} = A^{-1}\mathbf{b}$, $I\mathbf{u} = A^{-1}\mathbf{b}$, and $\mathbf{u} = A^{-1}\mathbf{b}$.

MATRIX OPERATIONS

■ Theorem 6:

- a. If A is an invertible matrix, then A^{-1} is invertible and

$$(A^{-1})^{-1} = A$$

- b. If A and B are $n \times n$ invertible matrices, then so is AB , and the inverse of AB is the product of the inverses of A and B in the reverse order. That is,

$$(AB)^{-1} = B^{-1}A^{-1}$$

- c. If A is an invertible matrix, then so is A^T , and the inverse of A^T is the transpose of A^{-1} . That is,

$$(A^T)^{-1} = (A^{-1})^T$$

MATRIX OPERATIONS

- **Proof:** To verify statement (a), find a matrix C such that

$$A^{-1}C = I \quad \text{and} \quad CA^{-1} = I$$

- These equations are satisfied with A in place of C . Hence A^{-1} is invertible, and A is its inverse.
- Next, to prove statement (b), compute:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

- A similar calculation shows that $(B^{-1}A^{-1})(AB) = I$.
- For statement (c), use Theorem 3(d), read from right to left, $(A^{-1})^T A^T = (AA^{-1})^T = I^T = I$.
- Similarly, $A^T (A^{-1})^T = I^T = I$.

ELEMENTARY MATRICES

- Hence A^T is invertible, and its inverse is $(A^{-1})^T$.
- The generalization of Theorem 6(b) is as follows:
The product of $n \times n$ invertible matrices is invertible, and the inverse is the product of their inverses in the reverse order.
- An invertible matrix A is row equivalent to an identity matrix, and we can find A^{-1} by *watching the row reduction of A to I* .
- An **elementary matrix** is one that is obtained by performing a single elementary row operation on an identity matrix.

ELEMENTARY MATRICES

■ **Example 5:** Let $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$, $E_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$,

$$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}, \quad A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Compute E_1A , E_2A , and E_3A , and describe how these products can be obtained by elementary row operations on A .

ELEMENTARY MATRICES

- **Solution:** Verify that

$$E_1A = \begin{bmatrix} a & b & c \\ d & e & f \\ g - 4a & h - 4b & i - 4c \end{bmatrix}, E_2A = \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix},$$

$$E_3A = \begin{bmatrix} a & b & c \\ d & e & f \\ 5g & 5h & 5i \end{bmatrix}.$$

- Addition of -4 times row 1 of A to row 3 produces E_1A .

ELEMENTARY MATRICES

- An interchange of rows 1 and 2 of A produces E_2A , and multiplication of row 3 of A by 5 produces E_3A .
- Left-multiplication (that is, multiplication on the left) by E_1 in Example 1 has the same effect on any $3 \times n$ matrix.
- Since $E_1 \cdot I = E_1$, we see that E_1 *itself* is produced by this same row operation on the identity.

ELEMENTARY MATRICES

- Example 5 illustrates the following general fact about elementary matrices.
- If an elementary row operation is performed on an $m \times n$ matrix A , the resulting matrix can be written as EA , where the $m \times m$ matrix E is created by performing the same row operation on I_m .
- Each elementary matrix E is invertible. The inverse of E is the elementary matrix of the same type that transforms E back into I .

ELEMENTARY MATRICES

- **Theorem 7:** An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n , and in this case, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .
- **Proof:** Suppose that A is invertible.
- Then, since the equation $A\mathbf{x} = \mathbf{b}$ has a solution for each $\mathbf{b}$ (Theorem 5), A has a pivot position in every row.
- Because A is square, the n pivot positions must be on the diagonal, which implies that the reduced echelon form of A is I_n . That is, $A \sim I_n$.

ELEMENTARY MATRICES

- Now suppose, conversely, that $A \sim I_n$.
- Then, since each step of the row reduction of A corresponds to left-multiplication by an elementary matrix, there exist elementary matrices $E_1, \dots, E_p$ such that $A \sim E_1 A \sim E_2 (E_1 A) \sim \dots \sim E_p (E_{p-1} \dots E_1 A) = I_n$
- That is,

$$(1) \quad E_p \dots E_1 A = I_n$$

- Since the product $E_p \dots E_1$ of invertible matrices is invertible, (1) leads to

$$(E_p \dots E_1)^{-1} (E_p \dots E_1) A = (E_p \dots E_1)^{-1} I_n$$

$$A = (E_p \dots E_1)^{-1}$$

ALGORITHM FOR FINDING A^{-1}

- Thus A is invertible, as it is the inverse of an invertible matrix (Theorem 6). Also,

$$A^{-1} = \left[(E_p \dots E_1)^{-1} \right]^{-1} = E_p \dots E_1.$$

- Then $A^{-1} = E_p \dots E_1 \cdot I_n$, which says that A^{-1} results from applying $E_1, \dots, E_p$ successively to I_n .
- This is the same sequence in (1) that reduced A to I_n .
- Row reduce the augmented matrix $[A \ I]$. If A is row equivalent to I , then $[A \ I]$ is row equivalent to $\begin{bmatrix} I & A^{-1} \end{bmatrix}$. Otherwise, A does not have an inverse.

ALGORITHM FOR FINDING A^{-1}

- **Example 2:** Find the inverse of the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}, \text{ if it exists.}$$

- **Solution:**

$$[A \quad I] = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix}$$

ALGORITHM FOR FINDING A^{-1}

$$\sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & -9/2 & 7 & -3/2 \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix}$$

ALGORITHM FOR FINDING A^{-1}

- Theorem 7 shows, since $A \sim I$, that A is invertible, and

$$A^{-1} = \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix}$$

- Now, check the final answer.

$$AA^{-1} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix} \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

ANOTHER VIEW OF MATRIX INVERSION

- It is not necessary to check that $A^{-1}A = I$ since A is invertible.
- Denote the columns of I_n by $\mathbf{e}_1, \dots, \mathbf{e}_n$.
- Then row reduction of $[A \quad I]$ to $[I \quad A^{-1}]$ can be viewed as the simultaneous solution of the n systems

$$A\mathbf{x} = \mathbf{e}_1, A\mathbf{x} = \mathbf{e}_2, \dots, A\mathbf{x} = \mathbf{e}_n \quad (2)$$

where the “augmented columns” of these systems have all been placed next to A to form

$$[A \quad \mathbf{e}_1 \quad \mathbf{e}_2 \quad \cdots \quad \mathbf{e}_n] = [A \quad I].$$

ANOTHER VIEW OF MATRIX INVERSION

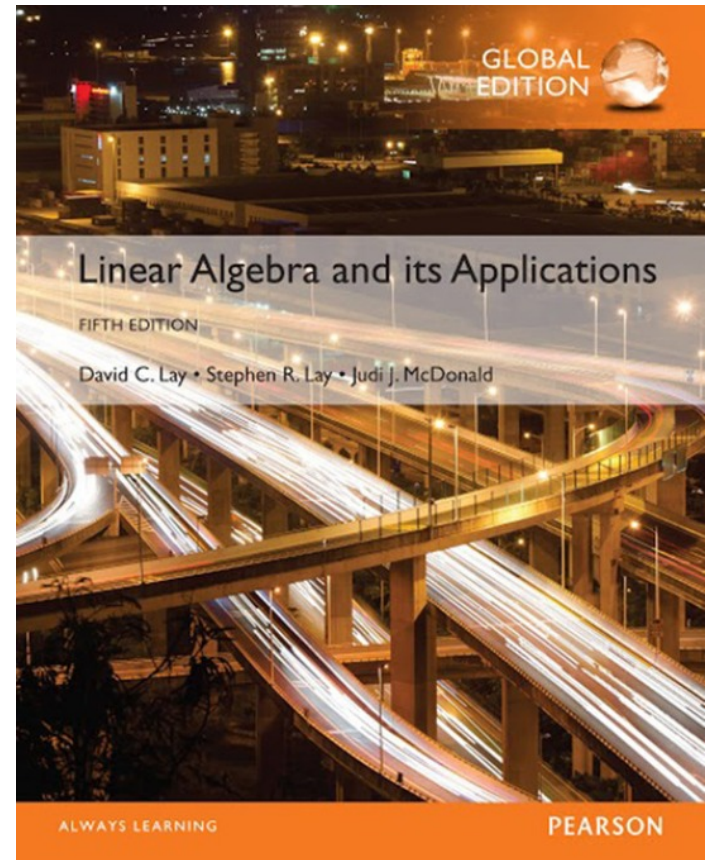
- The equation $AA^{-1} = I$ and the definition of matrix multiplication show that the columns of A^{-1} are precisely the solutions of the systems in (2).

2

Matrix Algebra

2.3

CHARACTERIZATIONS OF INVERTIBLE MATRICES



THE INVERTIBLE MATRIX THEOREM

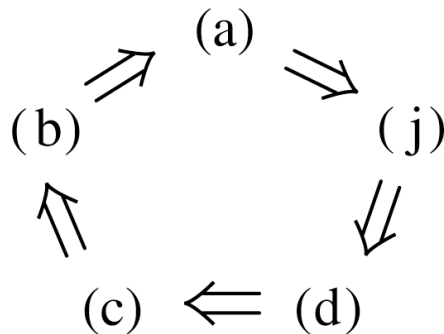
- **Theorem 8:** Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A , the statements are either all true or all false.
 - a. A is an invertible matrix.
 - b. A is row equivalent to the $n \times n$ identity matrix.
 - c. A has n pivot positions.
 - d. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
 - e. The columns of A form a linearly independent set.

THE INVERTIBLE MATRIX THEOREM

- f. The linear transformation $x \mapsto Ax$ is one-to-one.
- g. The equation $Ax = b$ has at least one solution for each b in $\mathbb{R}^n$.
- h. The columns of A span $\mathbb{R}^n$.
- i. The linear transformation $x \mapsto Ax$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- j. There is an $n \times n$ matrix C such that $CA = I$.
- k. There is an $n \times n$ matrix D such that $AD = I$.
- l. A^T is an invertible matrix.

THE INVERTIBLE MATRIX THEOREM

- First, we need some notation.
- If the truth of statement (a) always implies that statement (j) is true, we say that (a) *implies* (j) and write $(a) \Rightarrow (j)$.
- The proof will establish the “circle” of implications as shown in the following figure.



- If any one of these five statements is true, then so are the others.

THE INVERTIBLE MATRIX THEOREM

- Finally, the proof will link the remaining statements of the theorem to the statements in this circle.
- **Proof:** If statement (a) is true, then A^{-1} works for C in (j), so $(a) \Rightarrow (j)$.
- Next, $(j) \Rightarrow (d)$.
- Also, $(d) \Rightarrow (c)$.
- If A is square and has n pivot positions, then the pivots must lie on the main diagonal, in which case the reduced echelon form of A is I_n .
- Thus $(c) \Rightarrow (b)$.
- Also, $(b) \Rightarrow (a)$.

THE INVERTIBLE MATRIX THEOREM

- This completes the circle in the previous figure.
- Next, $(a) \Rightarrow (k)$ because A^{-1} works for D .
- Also, $(k) \Rightarrow (g)$ and $(g) \Rightarrow (a)$.
- So (k) and (g) are linked to the circle.
- Further, (g) , (h) , and (i) are equivalent for any matrix.
- Thus, (h) and (i) are linked through (g) to the circle.
- Since (d) is linked to the circle, so are (e) and (f) , because (d) , (e) , and (f) are all equivalent for *any* matrix A .
- Finally, $(a) \Rightarrow (l)$ and $(l) \Rightarrow (a)$.
- This completes the proof.

THE INVERTIBLE MATRIX THEOREM

- Theorem 8 could also be written as “The equation $A\mathbf{x} = \mathbf{b}$ has a *unique* solution for each $\mathbf{b}$ in $\mathbb{R}^n$.”
- This statement implies (b) and hence implies that A is invertible.
- The following fact follows from Theorem 8.
Let A and B be square matrices. If $AB = I$, then A and B are both invertible, with $B = A^{-1}$ and $A = B^{-1}$.
- The Invertible Matrix Theorem divides the set of all $n \times n$ matrices into two disjoint classes: the invertible (nonsingular) matrices, and the noninvertible (singular) matrices.

THE INVERTIBLE MATRIX THEOREM

- Each statement in the theorem describes a property of every $n \times n$ invertible matrix.
- The *negation* of a statement in the theorem describes a property of every $n \times n$ singular matrix.
- For instance, an $n \times n$ singular matrix is *not* row equivalent to I_n , does *not* have n pivot position, and has linearly *dependent* columns.

THE INVERTIBLE MATRIX THEOREM

- **Example 1:** Use the Invertible Matrix Theorem to decide if A is invertible:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & -2 \\ -5 & -1 & 9 \end{bmatrix}$$

- **Solution:**

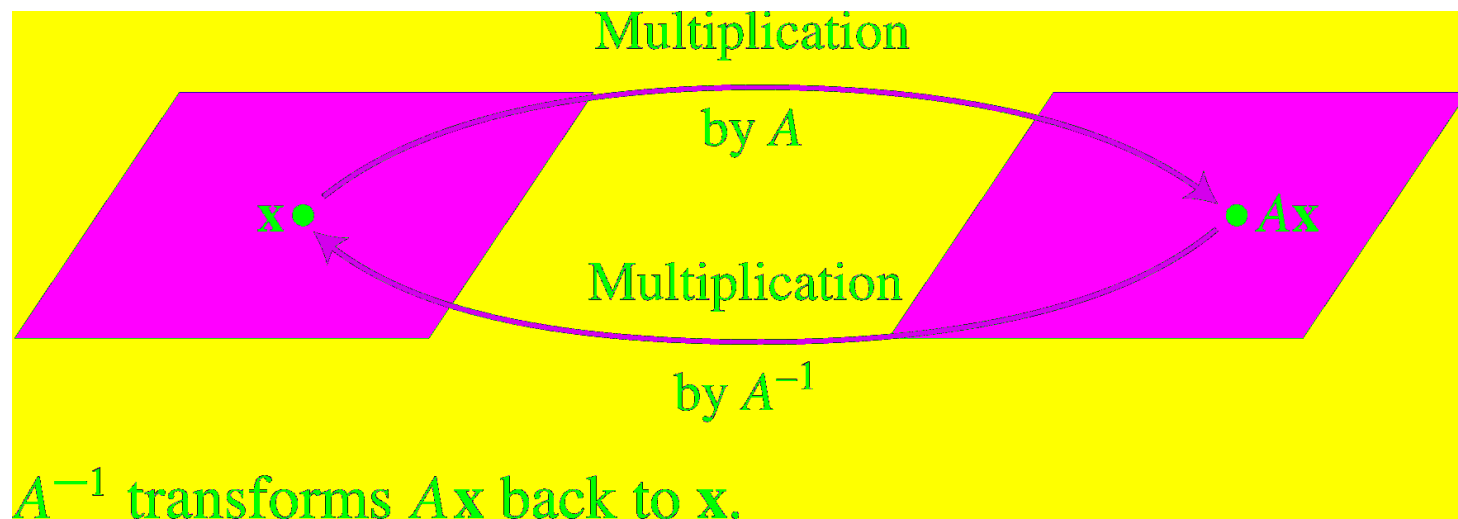
$$A \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

THE INVERTIBLE MATRIX THEOREM

- So A has three pivot positions and hence is invertible, by the Invertible Matrix Theorem, statement (c).
- The Invertible Matrix Theorem *applies only to square matrices*.
- For example, if the columns of a 4×3 matrix are linearly independent, we cannot use the Invertible Matrix Theorem to conclude anything about the existence or nonexistence of solutions of equation of the form $Ax = b$.

INVERTIBLE LINEAR TRANSFORMATIONS

- Matrix multiplication corresponds to composition of linear transformations.
- When a matrix A is invertible, the equation $A^{-1}Ax = x$ can be viewed as a statement about linear transformations. See the following figure.



INVERTIBLE LINEAR TRANSFORMATIONS

- A linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is said to be **invertible** if there exists a function $S: \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that

$$S(T(\mathbf{x})) = \mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^n \quad (1)$$

$$T(S(\mathbf{x})) = \mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^n \quad (2)$$

- **Theorem 9:** Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation and let A be the standard matrix for T . Then T is invertible if and only if A is an invertible matrix. In that case, the linear transformation S given by $S(\mathbf{x}) = A^{-1}\mathbf{x}$ is the unique function satisfying equation (1) and (2).

INVERTIBLE LINEAR TRANSFORMATIONS

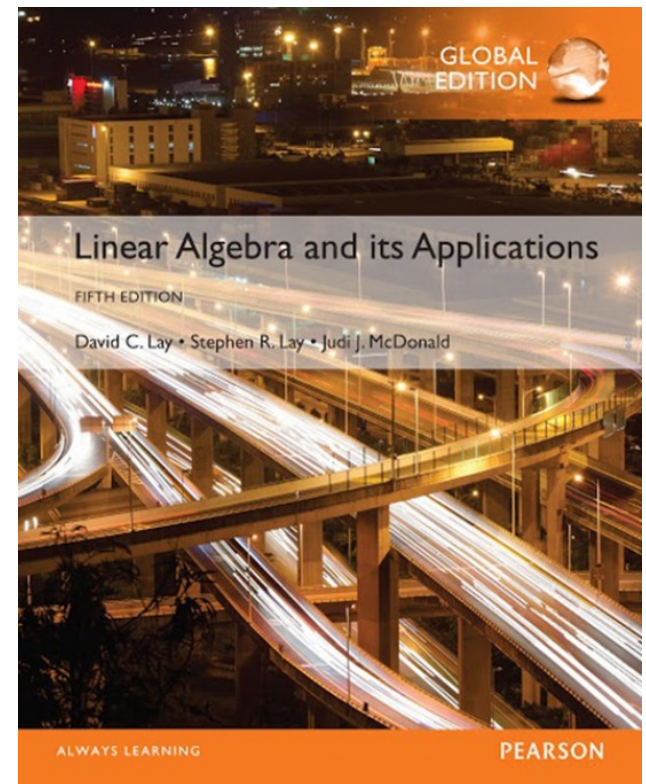
- **Proof:** Suppose that T is invertible.
- Then (2) shows that T is onto $\mathbb{R}^n$, for if $\mathbf{b}$ is in $\mathbb{R}^n$ and $\mathbf{x} = S(\mathbf{b})$, then $T(\mathbf{x}) = T(S(\mathbf{b})) = \mathbf{b}$, so each $\mathbf{b}$ is in the range of T .
- Thus A is invertible, by the Invertible Matrix Theorem, statement (i).
- Conversely, suppose that A is invertible, and let $S(\mathbf{x}) = A^{-1}\mathbf{x}$. Then, S is a linear transformation, and S satisfies (1) and (2).
- For instance, $S(T(\mathbf{x})) = S(A\mathbf{x}) = A^{-1}(A\mathbf{x}) = \mathbf{x}$.
- Thus, T is invertible.

2

Matrix Algebra

2.4

PARTITIONED MATRICES



PARTITIONED MATRICES

- A key feature of our work with matrices has been the ability to regard matrix A as a list of column vectors rather than just a rectangular array of numbers.
- This point of view has been so useful that we wish to consider other **partitions** of A , indicated by horizontal and vertical dividing rules, as in Example 1 on the next slide.

PARTITIONED MATRICES

- **Example 1** The matrix

$$A = \left[\begin{array}{ccc|cc|c} 3 & 0 & -1 & 5 & 9 & -2 \\ -5 & 2 & 4 & 0 & -3 & 1 \\ \hline -8 & -6 & 3 & 1 & 7 & -4 \end{array} \right]$$

- Can also be written as the 2×3 **partitioned** (or **block**) **matrix**

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}$$

- Whose entries are the *blocks* (or *submatrices*)

$$A_{11} = \begin{bmatrix} 3 & 0 & -1 \\ -5 & 2 & 4 \end{bmatrix}, \quad A_{12} = \begin{bmatrix} 5 & 9 \\ 0 & -3 \end{bmatrix}, \quad A_{13} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$A_{21} = \begin{bmatrix} -8 & -6 & 3 \end{bmatrix}, \quad A_{22} = \begin{bmatrix} 1 & 7 \end{bmatrix}, \quad A_{23} = \begin{bmatrix} -4 \end{bmatrix}$$

ADDITION AND SCALAR MULTIPLICATION

- If matrices A and B are the same size and are partitioned in exactly the same way, then it is natural to make the same partition of the ordinary matrix sum $A + B$.
- In this case, each block of $A + B$ is the (matrix) sum of the corresponding blocks of A and B .
- Multiplication of a partitioned matrix by a scalar is also computed block by block.

MULTIPLICATION OF PARTITIONED MATRICES

- Partitioned matrices can be multiplied by the usual row—column rule as if the block entries were scalars, provided that for a product AB , the column partition of A matches the row partition of B .

- **Example 3** Let

$$A = \left[\begin{array}{ccc|cc} 2 & -3 & 1 & 0 & -4 \\ 1 & 5 & -2 & 3 & -1 \\ 0 & -4 & -2 & 7 & -1 \end{array} \right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \left[\begin{array}{cc} 6 & 4 \\ -2 & 1 \\ -3 & 7 \\ -1 & 3 \\ 5 & 2 \end{array} \right] = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

- The 5 columns of A are partitioned into a set of 3 columns and then a set of 2 columns. The 5 rows of B are partitioned in the same way—into a set of 3 rows and then a set of 2 rows.

MULTIPLICATION OF PARTITIONED MATRICES

- We say that the partitions of A and B are conformable for block multiplication. It can be shown that the ordinary product AB can be written as

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} A_{11}B_1 + A_{12}B_2 \\ A_{21}B_1 + A_{22}B_2 \end{bmatrix} = \begin{bmatrix} -5 & 4 \\ -6 & 2 \\ \hline 2 & 1 \end{bmatrix}$$

- It is important for each smaller product in the expression for AB to be written with the submatrix from A on the left, since matrix multiplication is not commutative.

MULTIPLICATION OF PARTITIONED MATRICES

- For instance,

$$A_{11}B_1 = \begin{bmatrix} 2 & -3 & 1 \\ 1 & 5 & -2 \end{bmatrix} \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix}$$

$$A_{12}B_2 = \begin{bmatrix} 0 & -4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix}$$

- Hence the top block in AB is

$$A_{11}B_1 + A_{12}B_2 = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix} + \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix} = \begin{bmatrix} -5 & 4 \\ -6 & 2 \end{bmatrix}$$

MULTIPLICATION OF PARTITIONED MATRICES

- **Theorem 10:** Column—Row Expansion of AB

If A is $m \times n$ and B is $n \times p$, then

$$\begin{aligned} AB &= [\text{col}_1(A) \quad \text{col}_2(A) \quad \cdots \quad \text{col}_n(A)] \begin{bmatrix} \text{row}_1(B) \\ \text{row}_2(B) \\ \vdots \\ \text{row}_n(B) \end{bmatrix} \\ &= \text{col}_1(A) \text{row}_1(B) + \cdots + \text{col}_n(A) \text{row}_n(B) \end{aligned} \quad (1)$$

- **Proof** For each row index i and column index j , the (i, j) -entry in $\text{col}_k(A) \text{row}_k(B)$ is the product of a_{ik} from $\text{col}_k(A)$ and b_{kj} from $\text{row}_k(B)$.

MULTIPLICATION OF PARTITIONED MATRICES

- Hence the (i, j) -entry in the sum shown in equation (1) is

$$\begin{array}{ccccccc} a_{i1}b_{1j} & + & a_{i2}b_{2j} & + & \cdots & + & a_{in}b_{nj} \\ (k=1) & & (k=2) & & & & (k=n) \end{array}$$

- This sum is also the (i, j) -entry in AB , by the row—column rule.

INVERSES OF PARTITIONED MATRICES

- The next example illustrates calculations involving inverses and partitioned matrices.
- **Example 5** A matrix of the form

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$$

- Is said to be *block upper triangular*. Assume that A_{11} is $p \times p$, A_{22} is $q \times q$, and A is invertible. Find a formula for A^{-1} .

INVERSES OF PARTITIONED MATRICES

- **Solution** Denote A^{-1} by B and partition B so that

$$\begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} I_p & 0 \\ 0 & I_q \end{bmatrix} \quad (2)$$

- This matrix equation provides four equations that will lead to the unknown blocks $B_{11}, \dots, B_{22}$. Compute the product on the left side of equation (2), and equate each entry with the corresponding block in the identity matrix on the right.

INVERSES OF PARTITIONED MATRICES

- That is, set

$$A_{11}B_{11} + A_{12}B_{21} = Ip \quad (3)$$

$$A_{11}B_{12} + A_{12}B_{22} = 0 \quad (4)$$

$$A_{22}B_{21} = 0 \quad (5)$$

$$A_{22}B_{22} = Iq \quad (6)$$

- By itself, equation (6) does not show that A_{22} is invertible. However, since A_{22} is square, the Invertible Matrix Theorem and (6) together show that A_{22} is invertible and $B_{22} = A_{22}^{-1}$.

INVERSES OF PARTITIONED MATRICES

- Next, left-multiply both sides of (5) by A_{22}^{-1} and obtain

$$B_{21} = A_{22}^{-1} 0 = 0$$

- So that (3) simplifies to

$$A_{11}B_{11} + 0 = I_p$$

- Since A_{11} is square, this shows that A_{11} is invertible and $B_{11} = A_{11}^{-1}$. Finally, use these results with (4) to find that

$$A_{11}B_{12} = -A_{12}B_{22} = -A_{12}A_{22}^{-1} \quad \text{and} \quad B_{12} = -A_{11}^{-1}A_{12}A_{22}^{-1}$$

INVERSES OF PARTITIONED MATRICES

- Thus

$$A^{-1} = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}^{-1} = \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}$$

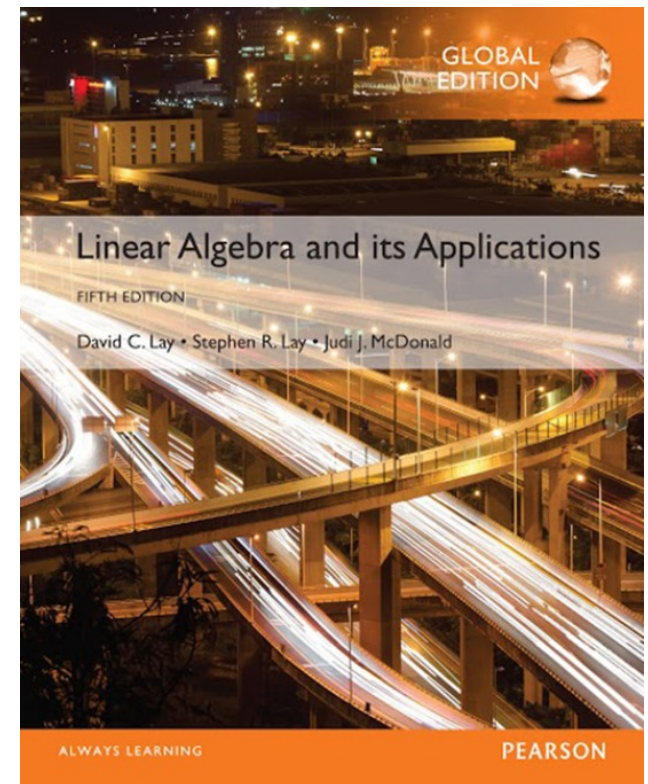
- A **block diagonal matrix** is a partitioned matrix with zero blocks off the main diagonal (of blocks). Such a matrix is invertible if and only if each block on the diagonal is invertible.

2

Matrix Algebra

2.5

MATRIX FACTORIZATIONS



MATRIX FACTORIZATIONS

- A *factorization* of a matrix A is an equation that expresses A as a product of two or more matrices.
- Whereas matrix multiplication involves a *synthesis* of data (combining the effects of two or more linear transformations into a single matrix), matrix factorization is an *analysis* of data.

THE LU FACTORIZATION

- The LU factorization, described on the next few slides, is motivated by the fairly common industrial and business problem of solving a sequence of equations, all with the same coefficient matrix:

$$Ax = b_1, \quad Ax = b_2, \dots, \quad Ax = bp \quad (1)$$

- When A is invertible, one could compute A^{-1} and then compute $A^{-1}b_1, A^{-1}b_2$, and so on.
- However, it is more efficient to solve the first equation in the sequence (1) by row reduction and obtain the LU factorization of A at the same time. Thereafter, the remaining equations in sequence (1) are solved with the LU factorization.

THE LU FACTORIZATION

- At first, assume that A is an $m \times n$ matrix that can be row reduced to echelon form, *without row interchanges*.
- Then A can be written in the form $A = LU$, where L is an $m \times m$ lower triangular matrix with 1's on the diagonal and U is an $m \times n$ echelon form of A .
- For instance, see Fig. 1 below. Such a factorization is called an **LU factorization** of A . The matrix L is invertible and is called a unit lower triangular matrix.

$$A = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_U$$

THE LU FACTORIZATION

- Before studying how to construct L and U , we should look at why they are so useful. When $A = LU$, the equation $Ax = b$ can be written as $L(Ux) = b$.
- Writing y for Ux , we can find x by solving the pair of equations

$$\begin{array}{l} Ly = b \\ Ux = y \end{array}$$

- First solve $Ly = b$ for y , and then solve $Ux = y$ for x . See Fig. 2 on the next slide. Each equation is easy to solve because L and U are triangular.

THE LU FACTORIZATION

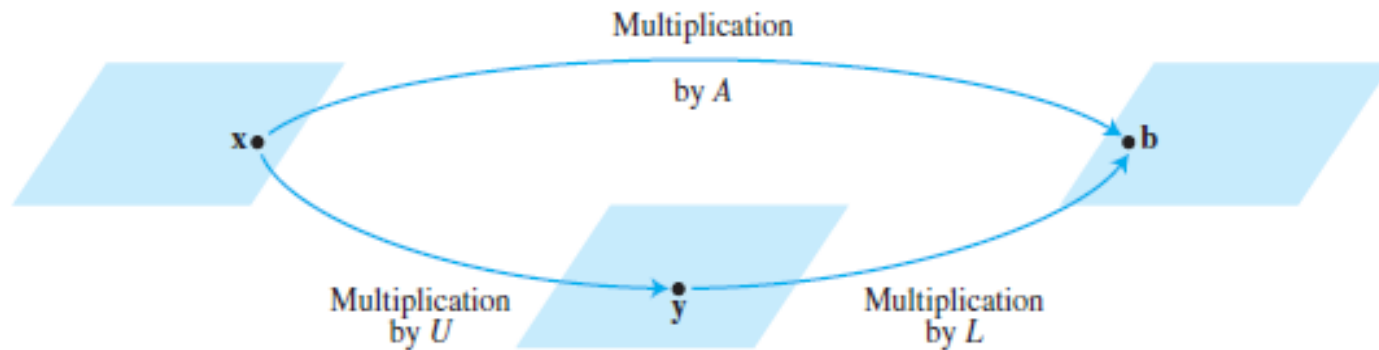


FIGURE 2 Factorization of the mapping $x \mapsto Ax$.

- **Example 1** It can be verified that

$$A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -5 & 1 & 0 \\ -3 & 8 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} = LU$$

- Use this factorization of A to solve $Ax=b$, where $b = \begin{bmatrix} -9 \\ 5 \\ 7 \\ 11 \end{bmatrix}$

THE LU FACTORIZATION

- **Solution** The solution of $Ly = b$ needs only 6 multiplications and 6 additions, because the arithmetic takes place only in column 5.

$$[L \quad \mathbf{b}] = \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ -1 & 1 & 0 & 0 & 5 \\ 2 & -5 & 1 & 0 & 7 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = [I \quad \mathbf{y}]$$

- Then, for $Ux = y$, the “backward” phase of row reduction requires 4 divisions, 6 multiplications, and 6 additions.

THE LU FACTORIZATION

- For instance, creating the zeros in column 4 of $[U \ y]$ requires 1 division in row 4 and 3 multiplication-addition pairs to add multiples of row 4 to the rows above.

$$[U \ y] = \begin{bmatrix} 3 & -7 & -2 & 2 & -9 \\ 0 & -2 & -1 & 2 & -4 \\ 0 & 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 3 \\ 4 \\ -6 \\ -1 \end{bmatrix}$$

- To find x requires 28 arithmetic operations, or “flops” (floating point operations), excluding the cost of finding L and U . In contrast, row reduction of $[A \ b]$ to $[I \ x]$ takes 62 operations.

AN LU FACTORIZATION ALGORITHM

- Suppose A can be reduced to an echelon form U using only row replacements that add a multiple of one row to another below it.
- In this case, there exist unit lower triangular elementary matrices $E_1, \dots, E_p$ such that

$$E_p \dots E_1 A = U$$

- Then (3)

$$A = (E_p \dots E_1)^{-1} U = LU$$

- where

$$L = (E_p \dots E_1)^{-1} \quad (4)$$

- It can be shown that products and inverses of unit lower triangular matrices are also unit lower triangular. Thus L is unit lower triangular.

AN LU FACTORIZATION ALGORITHM

- Note that row operations in equation (3), which reduce A to U , also reduce the L in equation (4) to I , because $E_p \dots E_1 L = (E_p \dots E_1)(E_p \dots E_1)^{-1} = I$. This observation is the key to *constructing* L .

Algorithm for an LU Factorization

1. Reduce A to an echelon form U by a sequence of row replacement operations, if possible.
2. Place entries in L such that the *same sequence of row operations* reduces L to I .

AN LU FACTORIZATION ALGORITHM

- Step 1 is not always possible, but when it is, the argument above shows that an LU factorization exists.
- Example 2 on the followings slides will show how to implement step 2. By construction, L will satisfy

$$(E_p \dots E_1)L = I$$

- using the same $E_p, \dots, E_1$ as in equation (3). Thus L will be invertible, by the Invertible Matrix Theorem, with $(E_p \dots E_1) = L^{-1}$. From (3), $L^{-1}A = U$, and $A = LU$. So step 2 will produce an acceptable L .

AN LU FACTORIZATION ALGORITHM

- **Example 2** Find an LU factorization of

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

- **Solution** Since A has four rows, L should be 4×4 . The first column of L is the first column of A divided by the top pivot entry:

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & & 1 & 0 \\ -3 & & & 1 \end{bmatrix}$$

AN LU FACTORIZATION ALGORITHM

- Compare the first columns of A and L . *The row operations that create zeros in the first column of A will also create zeros in the first column of L .*
- To make this same correspondence of row operations on A hold for the rest of L , watch a row reduction of A to an echelon form U . That is, *highlight the entries* in each matrix that are used to determine the sequence of row operations that transform A onto U .

$$\begin{aligned}
 A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix} &\sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix} = A_1 \\
 &\sim A_2 = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix} = U
 \end{aligned} \tag{5}$$

AN LU FACTORIZATION ALGORITHM

- The highlighted entries above determine the row reduction of A to U . At each pivot column, divide the highlighted entries by the pivot and place the result onto L :

$$\begin{array}{cccc}
 \begin{bmatrix} 2 \\ -4 \\ 2 \\ -6 \end{bmatrix} & \begin{bmatrix} 3 \\ -9 \\ 12 \end{bmatrix} & \begin{bmatrix} 2 \\ 4 \end{bmatrix} & [5] \\
 \div 2 & \div 3 & \div 2 & \div 5 \\
 \downarrow & \downarrow & \downarrow & \downarrow \\
 \begin{bmatrix} 1 & & & \\ -2 & 1 & & \\ 1 & -3 & 1 & \\ -3 & 4 & 2 & 1 \end{bmatrix} & \text{and } L = & \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}
 \end{array}$$

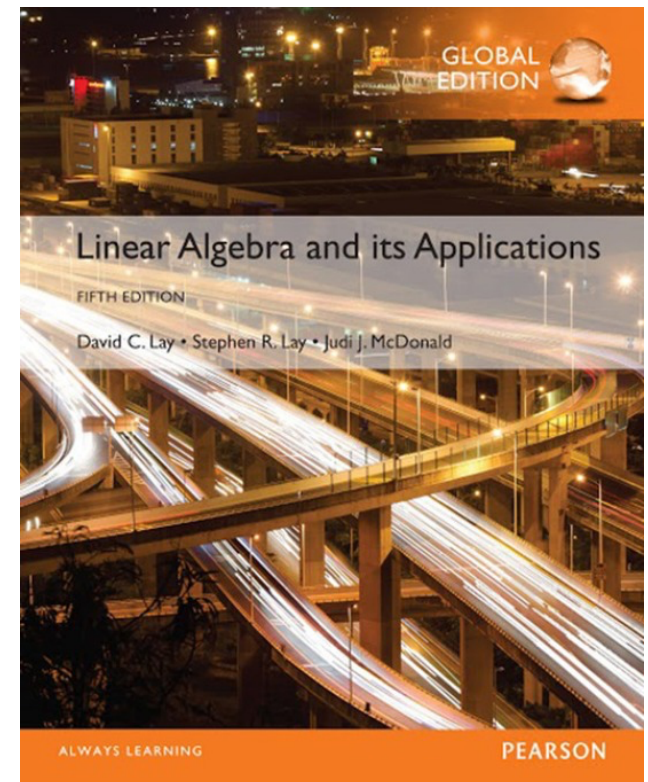
- An easy calculation verifies that this L and U satisfy $LU = A$.

2

Matrix Algebra

2.6

THE LEONTIEF INPUT- OUTPUT MODEL



THE LEONTIEF INPUT-OUTPUT MODEL

- Linear algebra played an essential role in the Nobel prize-winning work of Wassily Leontief.
- Suppose a nation's economy is divided into n sectors that produce goods or services, and let $\mathbf{x}$ be a **production vector** in $\mathbb{R}^n$ that lists the output of each sector for one year.
- Also, suppose another part of the economy (called the *open sector*) does not produce goods or services but only consumes them, and let $\mathbf{d}$ be a **final demand vector** (or **bill of final demands**) that lists the values of the goods and services demanded from the various sectors by the nonproductive part of the economy.

THE LEONTIEF INPUT-OUTPUT MODEL

- As the various sectors produce goods to meet consumer demand, the producers themselves create additional intermediate demand for goods they need as inputs for their own production.
- Leontief asked if there is a production level x such that the amounts produced (or “supplied”) will exactly balance the total demand for that production, so that

$$\left\{ \begin{array}{c} \text{amount} \\ \text{produced} \\ x \end{array} \right\} = \left\{ \begin{array}{c} \text{intermediate} \\ \text{demand} \end{array} \right\} + \left\{ \begin{array}{c} \text{final} \\ \text{demand} \\ d \end{array} \right\} \quad (1)$$

THE LEONTIEF INPUT-OUTPUT MODEL

- The basic assumption of Leontief's input-output model is that for each sector, there is a **unit consumption vector** in $\mathbb{R}^n$ that lists the inputs needed *per unit of output* of the sector.
- As a simple example, suppose the economy consists of three sectors—manufacturing, agriculture, and services—with unit consumption vectors c_1 , c_2 , and c_3 , as shown in the table that follows.

Purchased from:	Inputs Consumed per Unit of Output		
	Manufacturing	Agriculture	Services
Manufacturing	.50	.40	.20
Agriculture	.20	.30	.10
Services	.10	.10	.30
	↑	↑	↑
	c_1	c_2	c_3

THE LEONTIEF INPUT-OUTPUT MODEL

- **Example 1** What amounts will be consumed by the manufacturing sector if it decides to produce 100 units?

- **Solution** Compute

$$100c_1 = 100 \begin{bmatrix} .50 \\ .20 \\ .10 \end{bmatrix} = \begin{bmatrix} 50 \\ 20 \\ 10 \end{bmatrix}$$

- To produce 100 units, manufacturing will order (i.e., “demand”) and consume 50 units from other parts of the manufacturing sector, 20 units from agriculture, and 10 units from services.

THE LEONTIEF INPUT-OUTPUT MODEL

- If manufacturing decides to produce x_1 units of output, then x_1c_1 represents the *intermediate demands* of manufacturing, because the amounts in x_1c_1 will be consumed in the process of creating the x_1 units of output.
- Likewise, if x_2 and x_3 denote the planned outputs of the agriculture and services sectors, x_2c_2 and x_3c_3 list their corresponding intermediate demands.

THE LEONTIEF INPUT-OUTPUT MODEL

- The total intermediate demand from all three sectors is given by

$$\begin{aligned}\{\text{intermediate demand}\} &= x_1c_1 + x_2c_2 + x_3c_3 \\ &= Cx\end{aligned}\quad (2)$$

- where C is the consumption matrix $[c_1 \ c_2 \ c_3]$, namely,

$$C = \begin{bmatrix} .50 & .40 & .20 \\ .20 & .30 & .10 \\ .10 & .10 & .30 \end{bmatrix}\quad (3)$$

- Equations (1) and (2) yield Leontief's model.

$$\begin{array}{ccccc} \mathbf{x} & = & C\mathbf{x} & + & \mathbf{d} \\ \text{Amount} & & \text{Intermediate} & & \text{Final} \\ \text{produced} & & \text{demand} & & \text{Demand} \end{array}\quad (4)$$

THE LEONTIEF INPUT-OUTPUT MODEL

- Equation (4) may also be written as $I\mathbf{x} - C\mathbf{x} = \mathbf{d}$, or

$$(I - C)\mathbf{x} = \mathbf{d} \quad (5)$$

- Example 2** Consider the economy whose consumption matrix is given by (3). Suppose the final demand is 50 units for manufacturing, 30 units for agriculture, and 20 units for services. Find the production level x that will satisfy this demand.

- Solution** The coefficient matrix in (5) is

$$I - C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} .5 & .4 & .2 \\ .2 & .3 & .1 \\ .1 & .1 & .3 \end{bmatrix} = \begin{bmatrix} .5 & -.4 & -.2 \\ -.2 & .7 & -.1 \\ -.1 & -.1 & .7 \end{bmatrix}$$

THE LEONTIEF INPUT-OUTPUT MODEL

- To solve (5), row reduce the augmented matrix

$$\begin{bmatrix} .5 & -.4 & -.2 & 50 \\ -.2 & .7 & -.1 & 30 \\ -.1 & -.1 & .7 & 20 \end{bmatrix} \sim \begin{bmatrix} 5 & -4 & -2 & 500 \\ -2 & 7 & -1 & 300 \\ -1 & -1 & 7 & 200 \end{bmatrix} \sim \dots \sim \begin{bmatrix} 1 & 0 & 0 & 226 \\ 0 & 1 & 0 & 119 \\ 0 & 0 & 1 & 78 \end{bmatrix}$$

- The last column is rounded to the nearest whole unit.
Manufacturing must produce approximately 226 units,
agriculture 119 units, and services only 78 units.

THE LEONTIEF INPUT-OUTPUT MODEL

- In the theorem, the term **column sum** denotes the sum of the entries in a column of a matrix. Under ordinary circumstances, the column sums of a consumption matrix are less than 1 because a sector should require less than one unit's worth of inputs to produce one unit of output.
- **Theorem 11:** Let C be the consumption matrix for an economy, and let d be the final demand. If C and d have nonnegative entries and if each column sum of C is less than 1, then $(I - C)^{-1}$ exists and the production vector

$$\mathbf{x} = (I - C)^{-1}\mathbf{d}$$

has nonnegative entries and is the unique solution of

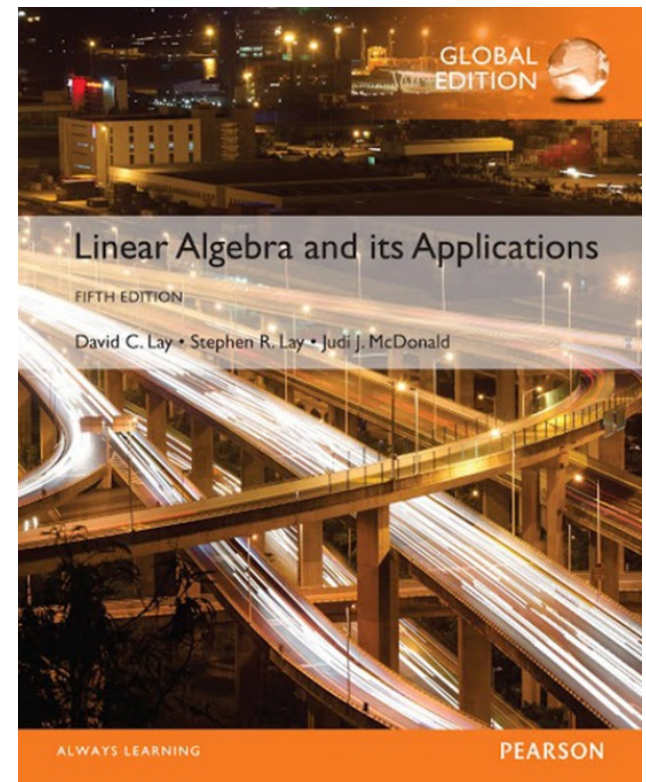
$$\mathbf{x} = C\mathbf{x} + \mathbf{d}$$

2

Matrix Algebra

2.8

SUBSPACES OF $\mathbb{R}^n$



SUBSPACES OF $\mathbb{R}^n$

- **Definition:** A **subspace** of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:
 - a) The zero vector is in H .
 - b) For each $\mathbf{u}$ and $\mathbf{v}$ in H , the sum $\mathbf{u} + \mathbf{v}$ is in H .
 - c) For each $\mathbf{u}$ in H and each scalar c , the vector $c\mathbf{u}$ is in H .

SUBSPACES OF $\mathbb{R}^n$

- A plane through the origin is the standard way to visualize the subspace in Example 1 on the next slide. See Fig. 1 below:

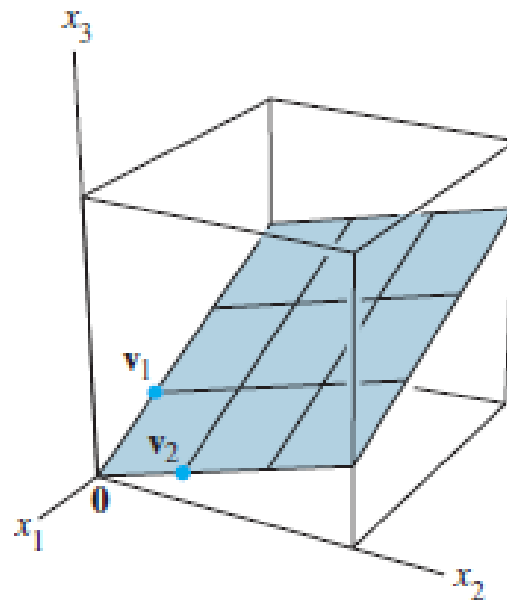


FIGURE 1

Span $\{v_1, v_2\}$ as a plane through the origin.

SUBSPACES OF $\mathbb{R}^n$

- **Example 1** If v_1 and v_2 are in $\mathbb{R}^n$ and $H = \text{Span}\{v_1, v_2\}$, then H is a subspace of $\mathbb{R}^n$. To verify this statement, note that the zero vector is in H (because $0v_1 + 0v_2$ is a linear combination of v_1 and v_2).

- Now take two arbitrary vectors in H , say,

$$u = s_1v_1 + s_2v_2 \quad \text{and} \quad v = t_1v_1 + t_2v_2$$

- Then

$$u + v = (s_1 + t_1)v_1 + (s_2 + t_2)v_2$$

- which shows that $u + v$ is a linear combination of v_1 and v_2 and hence is in H . Also, for any scalar c , the vector cu is in H , because $cu = c(s_1v_1 + s_2v_2) = cs_1(v_1) + cs_2(v_2)$.

COLUMN SPACE AND NULL SPACE OF A MATRIX

- **Definition:** The **column space** of a matrix A is the set $\text{Col } A$ of all linear combinations of the columns of A .
- If $A = [a_1 \dots a_n]$ with the columns of $\mathbb{R}^n$, then $\text{Col } A$ is the same as $\text{Span}\{a_1 \dots a_n\}$. Example 4 shows that the column space of an $m \times n$ matrix is a subspace of $\mathbb{R}^n$.
- **Example 4** Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$ and $b = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$.
Determine whether b is in the column space of A .

COLUMN SPACE AND NULL SPACE OF A MATRIX

- **Solution:** The vector $\mathbf{b}$ is a linear combination of the columns of A if and only if $\mathbf{b}$ can be written as $A\mathbf{x}$ for some $\mathbf{x}$, that is, if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution.

- Row reducing the augmented matrix $[A\mathbf{b}]$,

$$\begin{bmatrix} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & -2 & -6 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- We conclude that $A\mathbf{x} = \mathbf{b}$ is consistent and $\mathbf{b}$ is in $\text{Col } A$.

COLUMN SPACE AND NULL SPACE OF A MATRIX

- **Definition:** The **null space** of a matrix A is the set $\text{Nul}A$ of all solutions of the homogenous equation $Ax = 0$.
- **Theorem 12:** The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions of a system $Ax = 0$ of m homogenous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.
- **Proof:** The zero vector is in $\text{Nul}A$ (because $A0 = 0$). To show that $\text{Nul}A$ satisfies that other two properties required for a subspace, take any $\mathbf{u}$ and $\mathbf{v}$ in $\text{Nul}A$.

COLUMN SPACE AND NULL SPACE OF A MATRIX

- That is, suppose $A\mathbf{u} = 0$ and $A\mathbf{v} = 0$. Then, by a property of matrix multiplication,

$$A(u + v) = Au + Av = 0 + 0 = 0$$

- Thus $\mathbf{u} + \mathbf{v}$ satisfies $A = 0$, and so $\mathbf{u} + \mathbf{v}$ is in $\text{Nul}A$. Also, for any scalar c , $A(c\mathbf{u}) = c(A\mathbf{u}) = c(0) = 0$, which shows that $c\mathbf{u}$ is in $\text{Nul}A$.

BASIS FOR A SUBSPACE

- **Definition:** A **basis** for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H .
- **Example 5** The columns of an invertible $n \times n$ matrix form a basis for $\mathbb{R}^n$ because they are linearly independent and span $\mathbb{R}^n$, by the Invertible Matrix Theorem.

BASIS FOR A SUBSPACE

- One such matrix is the $n \times n$ identity matrix. Its columns are denoted by $e_1, \dots, e_n$:

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix},$$

- The set $\{e_1, \dots, e_n\}$ is called the **standard basis** for $\mathbb{R}^n$. See Fig. 3 on the next slide.

BASIS FOR A SUBSPACE

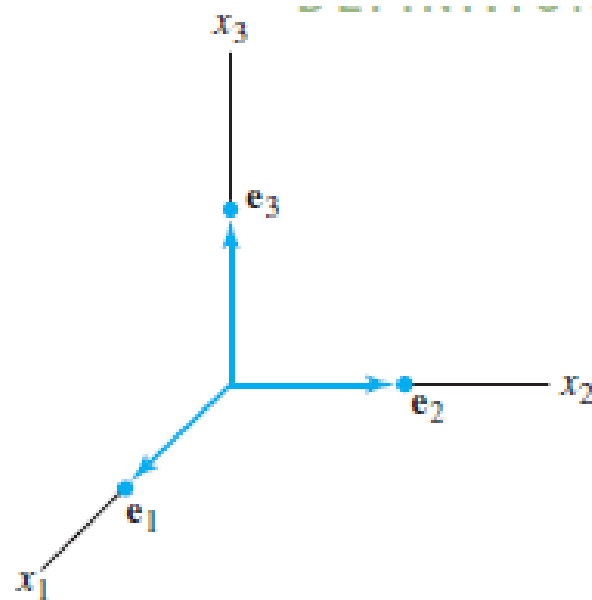


FIGURE 3

The standard basis for $\mathbb{R}^3$.

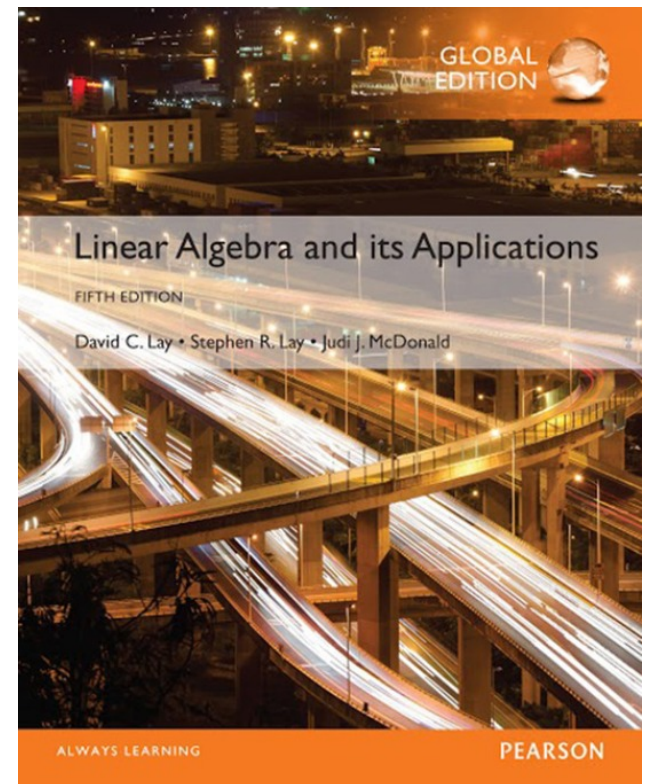
- **Theorem 13:** The pivot columns of a matrix A form a basis for the column space of A .

2

Matrix Algebra

2.9

DIMENSION AND RANK



COORDINATE SYSTEMS

- Suppose $\beta = \{b_1, \dots, b_p\}$ is a basis for H , and suppose a vector x in H can be generated in two ways, say,

$$x = c_1b_1 + \cdots + c_pb_p \text{ and } x = d_1b_1 + \cdots + d_pb_p \quad (1)$$

- Then, subtracting gives

$$0 = x - x = (c_1 - d_1)b_1 + \cdots + (c_p - d_p)b_p \quad (2)$$

- Since β is linearly independent, the weights in (2) must all be zero. That is, $c_j = d_j$ for $1 \leq j \leq p$, which shows that the two representations in (1) are actually the same.

COORDINATE SYSTEMS

- **Definition:** Suppose the set $\beta = \{b_1, \dots, b_p\}$ is a basis for a subspace H . For each x in H , the **coordinates of x relative to the basis β** are the weights $c_1, \dots, c_p$ such that $x = c_1b_1 + \dots + c_pb_p$, and the vector in $\mathbb{R}^p$

$$[x]_{\beta} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$$

- is called the **coordinate vector of x (relative to β)** or the **β -coordinate vector of x** .

COORDINATE SYSTEMS

- **Example 1** Let $v_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$, $v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $x = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$, and $\beta = \{v_1, v_2\}$. Then β is a basis for $H = \text{Span}\{v_1, v_2\}$ because v_1 and v_2 are linearly independent. Determine if x is in H , and if it is, find the coordinate vector of x relative to β .
- **Solution** If x is in H , then the following vector equation is consistent:

$$c_1 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

COORDINATE SYSTEMS

- The scalars c_1 and c_2 , if they exist, are the β -coordinates of $\mathbf{x}$. Row operations show that

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

- Thus $c_1 = 2$, $c_2 = 3$ and $[\mathbf{x}]_\beta = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. The basis β determines a “coordinate system” on H , which can be visualized by the grid shown in Fig. 1 on the next slide.

COORDINATE SYSTEMS

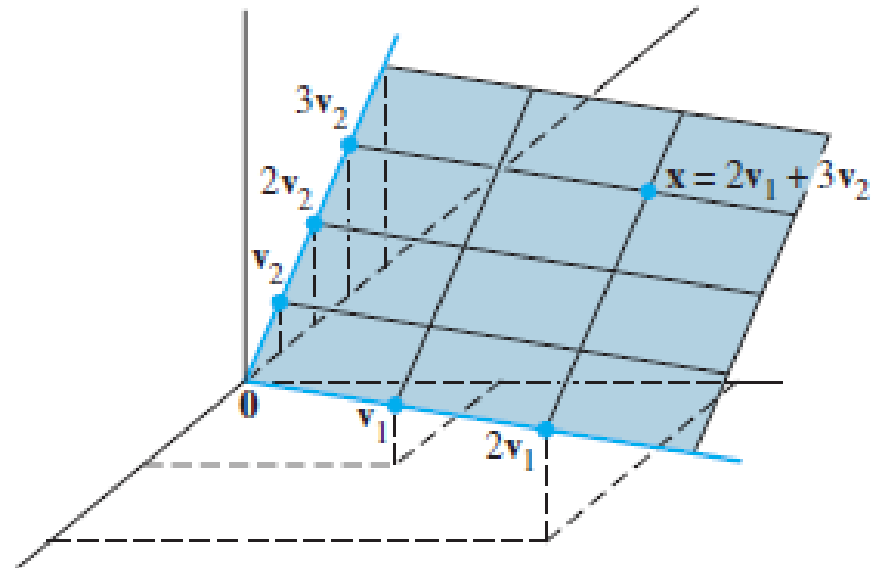


FIGURE 1 A coordinate system on a plane H in $\mathbb{R}^3$.

THE DIMENSION OF A SUBSPACE

- **Definition:** The **dimension** of a nonzero subspace H , denoted by $\dim H$, is the number of vectors in any basis for H . The dimension of the zero subspace $\{0\}$ is defined to be zero.
- **Definition:** The **rank** of a matrix A , denoted by $\text{rank } A$, is the dimension of the column space of A .

THE DIMENSION OF A SUBSPACE

- **Example 3** Determine the rank of the matrix

$$A \sim \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & -6 & 4 & 14 & -20 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix}$$

- **Solution** Reduce A to echelon form:

$$A \sim \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & -6 & 4 & 14 & -20 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix} \sim \dots \sim \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 4 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivot columns 

- The matrix A has 3 pivot columns, so $\text{rank } A = 3$.

THE DIMENSION OF A SUBSPACE

- **Theorem 14** If a matrix A has n columns, then $\text{rank } A + \dim \text{Nul}A = n$.
- **Theorem 15** Let H be a p -dimensional subspace of $\mathbb{R}^n$. Any linearly independent set of exactly p elements in H is automatically a basis for H . Also, any set of p elements of H that spans H is automatically a basis for H .

RANK AND THE INVERTIBLE MATRIX THEOREM

- **The Invertible Theorem (continued)** Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.
 - m. The columns of A form a basis of $\mathbb{R}^n$.
 - n. $\text{Col } A = \mathbb{R}^n$
 - o. $\dim \text{Col } A = n$
 - p. $\text{rank } A = n$
 - q. $\text{Nul } A = \{0\}$
 - r. $\dim \text{Nul } A = 0$

RANK AND THE INVERTIBLE MATRIX THEOREM

- **Proof** Statement (m) is logically equivalent to statements (e) and (h) regarding linear independence and spanning. The other five statements are linked to the earlier ones of the theorem by the following chain of almost trivial implications:
$$(g) \Rightarrow (n) \Rightarrow (o) \Rightarrow (p) \Rightarrow (r) \Rightarrow (q) \Rightarrow (d)$$
- Statement (g), which says that the equation $Ax = b$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$, implies statement (n), because $\text{Col } A$ is precisely the set of all $\mathbf{b}$ such that the equation $Ax = b$ is consistent.

RANK AND THE INVERTIBLE MATRIX THEOREM

- The implications $(n) \Rightarrow (o) \Rightarrow (p)$ follow from the definitions of *dimension* and *rank*.
- If the rank of A is n , the number of columns of A , then $\dim \text{Nul}A = 0$, by the Rank Theorem, and so $\text{Nul}A = \{0\}$. Thus $(p) \Rightarrow (r) \Rightarrow (q)$.
- Also, statement (q) implies that the equation $Ax = 0$ has only the trivial solution, which is statement (d).
- Since statements (d) and (g) are already known to be equivalent to the statement that A is invertible, the proof is complete.